

Interactions in regression analysis

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Reading and assignments in preparation of this class

In preparation for class, please read *Regression and other stories* (ROS), Ch. 10 (10.1-10.5). **You should plan for ~3h of reading and summarizing your questions in preparation for this week of the class.**

About this document

This PDF is generated from an Quarto markdown document. If you're interested in understanding the code that generated the examples presented below, you can [clone the repo](#). Both the PDF and the R markdown that the PDF is generated from are in the `scripts/` folder. You can go through the document line by line, or make changes and 'knit' a new PDF with the press of a button. You can also submit corrections or suggestions for improvements as a pull request to the GitHub repository.

The data for this document

This tutorial uses the same data introduced in the [tutorial on factor coding](#).

Interactions between factors and continuous predictors

We expand the analysis developed in the factor coding tutorial further by removing the assumption that the effects of `Condition` and `DiffusionConstant` are additive. To this end, we are adding a new predictor to the model that represents the interaction between `Condition` and `DiffusionConstant`.¹

For the remainder of this tutorial, we focus on the type of planned analysis typical for (or at least the idea of) experimental psychology. For that reason, we do not discuss considerations about when we should even consider an interaction (but see, for example, Harrell, 2001 for insightful discussion, as well as Gelman & Hill, 2007, p. 36).²

For the case of one continuous and one categorical predictor, the geometric interpretation of the model is that we now allow the two lines (corresponding to the continuous predictor's effect at the two levels of the categorical predictor) to not be parallel. And **the question of whether the interaction is *significant* becomes the question whether the difference in the slope of the two lines is statistically different from zero.**

We can ask this question by adding a new predictor to the model that is the *product* of the two predictors (cf. Gelman & Hill, 2007, p. 34-36; James et al., 2013, p. 87-90). We then ask whether this new predictor has a non-zero effect, i.e., we ask whether the coefficient for this new predictor is different from zero. In R, the regression formula for this model is written as (where the colon is the interaction operator):

```
Threshold ~ 1 + Condition + DiffusionConstant + Condition:DiffusionConstant
```

which essentially runs the following model (where `I` is the identity operator, which return `x` for `x`):

```
Threshold ~ 1 + Condition + DiffusionConstant + I(Condition  
* DiffusionConstant)
```

or shorter (where `X1 * X2` is a shorthand for the full-factorial combination of `X1` and `X2`):

```
Threshold ~ 1 + Condition*DiffusionConstant
```

¹ The test of whether two effects are additive is not to be confused with the assumption of the LM that all of the *terms* in the model—i.e., all the $x_k\beta_k$ —have additive effects. The additivity assumption of the model still holds: just as we assumed before that the effects of `Condition` and `DiffusionConstant` are additive, we now assume that the effects of `Condition`, `DiffusionConstant`, and their interaction are additive.

² The short of those considerations is that introducing additional complexity (degrees of freedom) increases the risk of overfitting to the data, and with it the risk of spurious/unreliable conclusions. It is thus advisable to include additional complexity in the model only if it is motivated theoretically, by previous work, or other *a priori* considerations—e.g., the design of the experiment. Another specific recommendation you'll see often is to only include interactions if all their components are also included in the model, unless there is a good theoretical reason to do otherwise. These recommendations are not specific to the LM.

Just like, e.g., factor coding implicit creates an additional numerical variable that encodes the factor's information, the interaction operator implicitly creates a new variable in our data that is the product of the (numerically coded) factor `Condition` and the continuous predictor `DiffusionConstant`. To see this, we again look at the model matrix:

With this change in contrast, the model matrix also changes:

```
knitr::kable(
  model.matrix(~ 1 + Condition * DiffusionConstant, data = d),
  format = "latex",
  booktabs = TRUE
) %>%
  kableExtra::kable_styling(
    latex_options = "HOLD_position",
    font_size = 8)
```

(Intercept)	ConditionCrowded vs. mean	DiffusionConstant	ConditionCrowded vs. mean:DiffusionConstant
1	-0.5	11.123342	-5.561671
1	-0.5	4.893119	-2.446560
1	-0.5	18.518145	-9.259072
1	-0.5	15.609822	-7.804911
1	-0.5	7.069902	-3.534951
1	-0.5	23.487701	-11.743851
1	-0.5	10.750472	-5.375236
1	-0.5	18.541809	-9.270905
1	-0.5	14.479893	-7.239946
1	-0.5	14.243763	-7.121881
1	0.5	15.638986	7.819493
1	0.5	5.644884	2.822442
1	0.5	39.431606	19.715803
1	0.5	14.408659	7.204330
1	0.5	6.249123	3.124561
1	0.5	30.619533	15.309767
1	0.5	25.096705	12.548352
1	0.5	15.031910	7.515955
1	0.5	11.302198	5.651099
1	0.5	17.261093	8.630547

Do the effects of condition and diffusion constant interact?

Now we can fit the new LM with the interaction, and look at the summary of results. We will use the centered diffusion constant for this model—i.e., we'll fit the model:

```
Threshold ~ 1 + Condition * DiffusionConstant_c
```

For this model, we can refer to the effect of condition as the “*main effect*” of condition (the overall, or mean, effect of condition, marginalizing over the effects of diffusion constant). If we didn’t center the diffusion constant—and more generally, if there were other uncentered variables in the model—the effect of condition would not be conventionally referred to as main effect.

Call:

```
lm(formula = Threshold ~ 1 + Condition * DiffusionConstant_c,
    data = d)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-0.30964 -0.14039 -0.05505  0.10824  0.45666
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.804444	0.047022	38.374	< 2e-
16 ***				
ConditionCrowded vs. mean	0.441727	0.094044	4.697	0.000242 ***
DiffusionConstant_c	0.032167	0.006725	4.783	0.000203 ***
ConditionCrowded vs. mean:DiffusionConstant_c	0.008665	0.013449	0.644	0.528534

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2006 on 16 degrees of freedom

Multiple R-squared: 0.835, Adjusted R-squared: 0.8041

F-statistic: 26.99 on 3 and 16 DF, p-value: 1.695e-06

The first thing we can notice is that the R^2 of the model has barely increased. In other words, adding the interaction doesn’t explain much more variance in the outcome. Indeed, the adjusted R^2 (which is the R^2 corrected for the complexity of the model) has *decreased*. With this in mind, it is not surprising that the interaction does not have a significant effect. In short, we cannot reject the null hypothesis that the effect of the diffusion constant on the threshold is identical for crowded and uncrowded condition. This is also in line with a visualization of the data and the model:

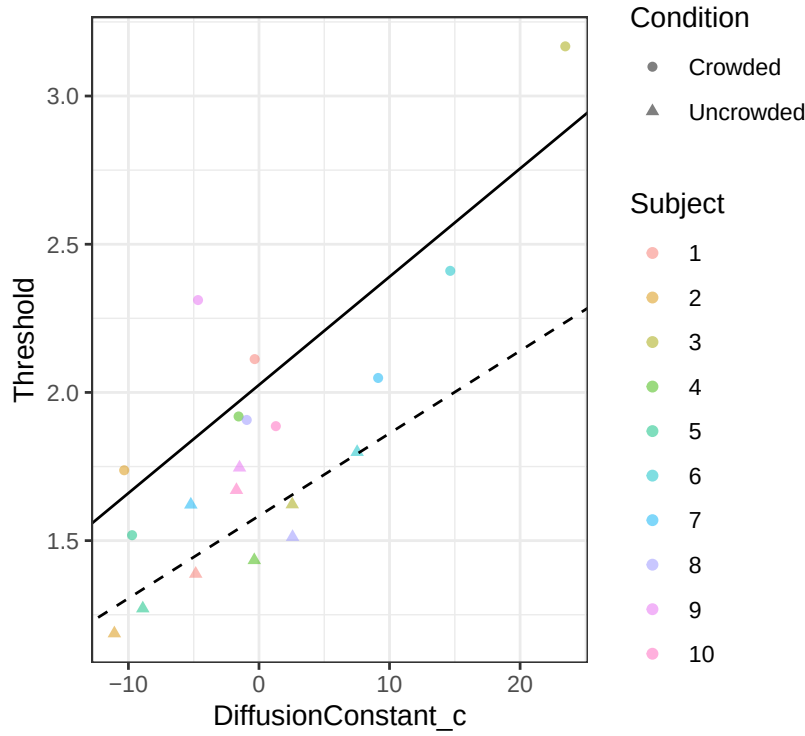


Figure 1: Size (in arcminutes) at which threshold performance (62.5% correct) was reached by diffusion constant (centered) and crowding condition. Each color shows a separate subject. Black lines show the predicted outcome of a linear model fit to the data.

In the LM with the interaction, the main effect of diffusion constant is the *average* effect of the diffusion constant across the two conditions (because we effect-coded and thus centered condition). The main effect of condition is the *average* across all values of the diffusion constant (because we centered the diffusion constant, so that the effect of condition is the predicted distance between the lines when the centered diffusion constant is zero, i.e., at the mean of the diffusion constant). The interaction tells us how much the slope of the diffusion constant differs between the conditions. Specifically, the slope of the diffusion constant for the crowded condition is predicted to be $+ 0.008665 * 0.05 = + 0.0043325$ larger than the main effect of the diffusion constant ($0.032167 + 0.0043325$), whereas the slope of the diffusion constant for the uncrowded condition is predicted to be $+ 0.008665 * -0.05 = - 0.0043325$ smaller than the main effect of the diffusion constant ($0.032167 - 0.0043325$).

Note also that the intercept estimate in the interaction model is not identical to the one from the combined model. This seems to contradict what we learned above, that the intercept estimate is the model's prediction for the grand mean if all predictors are centered. Note, however, that we did not center the interaction and, indeed, the mean of the product of condition and DiffusionConstant is *not* 0, but rather 1.049.³

Prepare for class

1. On Figure 3, draw a line segment that correspond to the intercept. I.e., where along the x-axis does the intercept make a prediction (a vertical line segment) and from where to where along y should the line segment go?
2. On Figure 3, draw a line segment that correspond to the effect of condition.
3. Try to calculate the slope for the crowded condition from the model output shown above. If you're not sure how to do it, discuss with your peers and post your questions/conclusions on Slack.
4. What's a geometric interpretation of an interaction between two *continuous* predictors (x1 and x2)? For this it might be helpful to first think about the geometric interpretation of two additive continuous predictors (a plane over x1 and x2, the height of which is given along the third axis, y)
5. True or false? Since the interaction does not have a significant effect, we can safely remove it from the model.

³ Since the diffusion constant is not a variable that we controlled directly by our design, it can have different means for the two conditions even after centering it, and indeed it does (centering only guarantees that the *overall* mean of the diffusion constant is 0): In other words, the diffusion constant is itself affected by the condition and, as a consequence, condition and the diffusion constants are correlated. Here we don't explore this further, but we will return to questions about collinearity later in the semester.

Discussion questions for class

6. True or false? Now that we know that the slope of diffusion constant does not significantly differ between the two crowdedness conditions, and given that the effect of diffusion constant was significant, we can conclude that both conditions exhibit a significant effect of the diffusion constant.
7. True or false? If we had found a significant interaction between condition and diffusion constant, we could have concluded that the effect of conclusion constant goes in opposite directions for the two conditions?
8. When we fit the same interaction model with the uncentered diffusion constant instead, we get a seemingly conflicting result, where condition does not longer have a significant effect. Why is that the case? Is this result really conflicting? (The figure might help. Hint: draw the line segments corresponding to the effect of condition in this new interaction model.)

```

Call:
lm(formula = Threshold ~ 1 + Condition * DiffusionConstant, data = d)

Residuals:
    Min       1Q   Median       3Q      Max
-0.30964 -0.14039 -0.05505  0.10824  0.45666

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.290728   0.109636   11.773 2.71e-
09 ***
ConditionCrowded vs. mean    0.303347   0.219271    1.383 0.185531
DiffusionConstant    0.032167   0.006725    4.783 0.000203 ***
ConditionCrowded vs. mean:DiffusionConstant 0.008665   0.013449    0.644 0.528534
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2006 on 16 degrees of freedom
Multiple R-squared:  0.835, Adjusted R-squared:  0.8041
F-statistic: 26.99 on 3 and 16 DF, p-value: 1.695e-06

```

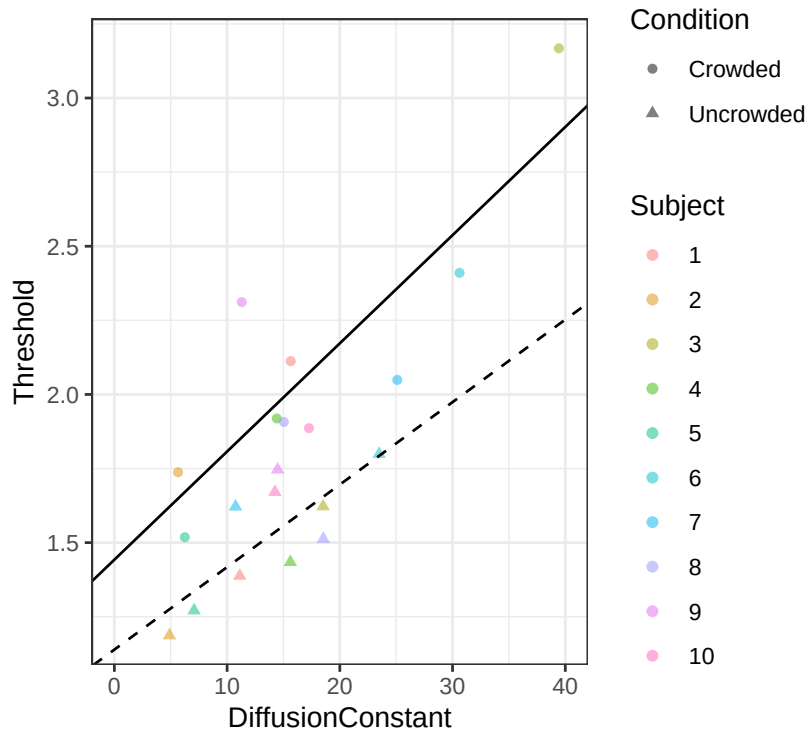


Figure 2: Size (in arcminutes) at which threshold performance (62.5% correct) was reached by diffusion constant (not-centered) and crowding condition. Each color shows a separate subject. Black lines show the predicted outcome of a linear model fit to the data.

1. Are the predicted/fitted values from this model different from the first interaction model we fit above?

Simple effects

When we have interactions in our model, we might also want to report the *simple effects*. For example, for the two-way interaction between condition and diffusion constant, we might want to report the simple effect of the diffusion constant for each condition. That's necessary because a significant interaction between two predictors x_1 and x_2 only tells us that the effect of x_1 on y differs depending on x_2 (or vice versa, that the effect of x_2 on y depends on the value of x_1). For example, for the present case a significant interaction would have indicated that the slope of diffusion constant differs between the two crowdedness conditions.

However, a significant interaction does *not* easily tell us *how* the interacting effects depends on each other (it actually does, but not in a way that's very easy to read from the model). For example, for the present case a significant interaction could indicate that there is a significant positive relation between the diffusion constant and threshold in one condition and a significant negative relation in the other condition; but it could also indicate that there is a significant positive relation in one condition and a non-significant relation in the other; or even that the relation is significant and positive in both conditions but the size of the effect of diffusion constant differs between condition (and so on). To address which of those scenarios holds, it is necessary to conduct additional analyses. The standard approach to that is called simple effect analyses.

In this approach, we do not split the data into the two conditions and fit separate models (`Threshold ~ 1 + DiffusionConstant`) to it. Rather, we use all of the data we have and simple re-parameterize/re-code the same LM we have already fit to read out the simple effects of `DiffusionConstant` at each level of condition. In R, this can be done with the convenient embedding operator `/`. Here's the model matrix resulting from this operator does for the present case:

```
knitr::kable(
  model.matrix(~ 1 + Condition / DiffusionConstant_c, data = d),
  format = "latex",
  booktabs = TRUE
) %>%
  kableExtra::kable_styling(
    latex_options = "HOLD_position",
    font_size = 8)
```

(Intercept)	ConditionCrowded vs. mean	ConditionCrowded:DiffusionConstant_c	ConditionUncrowded:DiffusionConstant_c
1	-0.5	0.0000000	-4.846791
1	-0.5	0.0000000	-11.077014
1	-0.5	0.0000000	2.548011
1	-0.5	0.0000000	-0.360311
1	-0.5	0.0000000	-8.900231
1	-0.5	0.0000000	7.517568
1	-0.5	0.0000000	-5.219661
1	-0.5	0.0000000	2.571676
1	-0.5	0.0000000	-1.490241
1	-0.5	0.0000000	-1.726370
1	0.5	-0.3311476	0.000000
1	0.5	-10.3252492	0.000000
1	0.5	23.4614730	0.000000
1	0.5	-1.5614743	0.000000
1	0.5	-9.7210102	0.000000
1	0.5	14.6494003	0.000000
1	0.5	9.1265712	0.000000
1	0.5	-0.9382233	0.000000
1	0.5	-4.6679358	0.000000
1	0.5	1.2909599	0.000000

and here is the resulting LM:

```
Call:
lm(formula = Threshold ~ 1 + Condition/DiffusionConstant_c, data = d)

Residuals:
    Min       1Q   Median       3Q      Max
-0.30964 -0.14039 -0.05505  0.10824  0.45666

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.804444   0.047022  38.374 < 2e-
16 ***
ConditionCrowded vs. mean      0.441727   0.094044   4.697 0.000242 ***
ConditionCrowded:DiffusionConstant_c  0.036500   0.006243   5.846 2.48e-
05 ***
ConditionUncrowded:DiffusionConstant_c 0.027835   0.011912   2.337 0.032790 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2006 on 16 degrees of freedom
Multiple R-squared:  0.835, Adjusted R-squared:  0.8041
F-statistic: 26.99 on 3 and 16 DF, p-value: 1.695e-06
```

This re-parameterization does not change the predicted/fitted responses of the model (not shown here) or the model fit (R^2 , F-statistic, etc.; shown). But we now are provided with the effect of the diffusion constant at each level of the condition variable. In this case, we see

that the effect of diffusion constant on the threshold is significant and positive in both conditions.

Writing up the analysis results

We analyzed the data with a linear regression, using the function `lm` from the `base` package (citation with version) of the software R (citation with version). We calculated each subject's mean thresholds for the crowded and uncrowded condition. These 20 threshold values were regressed against condition (effect-coded: $.5 = \text{crowded}$ vs. $-.5 = \text{uncrowded}$), the diffusion constant (**centered**), and **their interaction**. We found a statistically significant **main** effect of condition ($\hat{\beta} = .442, t = 4.697, p < .01$), so that subjects reach threshold performance at larger sizes in the crowded condition, compared to the uncrowded condition. We also found a statistically significant effect of the diffusion constant ($\hat{\beta} = .037, t = 5.846, p < .01$), so that larger diffusion constants required larger sizes to achieve threshold performance. **The interaction was not significant** ($\hat{\beta} = 0.009, p > .5$).⁴

⁴ Note that we usually don't report the simple effects unless the interaction was significant.

Discussion questions for class

In your favorite statistics program, try to fit a Bayesian linear regression with the interaction between `Condition` and `DiffusionConstant_c`.